The complex relationship between the tunes and pragmatics of Greek wh-questions

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Revised version of paper to appear in Eckardt, Regine, George Walkden & Nicole Dehé (eds.), *The Handbook of Noncanonical Questions*. Oxford University Press.

1. Introduction

The relationship between intonation and meaning has always been of interest (see Westera, Goodhue & Gussenhoven 2020, for a review). Intonation meaning has been examined both from the perspective of syntax, semantics, and pragmatics, and the perspective of phonetics and phonology. Approaches in the former mold often assume that tunes are related to specific communicative functions, such as indicating information structure distinctions (e.g., Steedman, 2000; Krifka, 2008; Büring, 2009). Phonetic and phonological approaches vary considerably in their attention to intonation meaning, with some models, such as the IPO ('t Hart, Collier, & Cohen, 1990) not being concerned with meaning at all, while others concentrate on specific topics, such as focus (e.g., Xu, 2005).

The present paper addresses intonation meaning in wh-questions in the context of the autosegmental-metrical framework of intonational phonology (henceforth AM; Pierrehumbert, 1980; Ladd, 2008; Arvaniti, 2022) and its compositional approach to intonation meaning. Briefly, in AM, tunes are composed of a string of low (L) and H (high) autosegments, known as *pitch accents* when associated with metrical heads, and as *edge tones* when associated with phrasal boundaries. Critically, these elements are also morphemes, combing form and meaning. Each intonation morpheme contributes independently to the overall meaning of the tune, which is then interpreted in context and in combination with syntax and lexical choices (Pierrehumbert & Hirschberg, 1990). Recent research supports this approach and shows that the interpretation of intonation meaning is probabilistic (Calhoun, 2010; Im, Cole & Baumann, 2018; Kurumada & Roettger, 2022). Further, since tunes are interpreted in context, terms such as *question intonation* or *focus intonation* are useful shorthands but should not be interpreted to mean that all questions or focused elements are produced with one tune that is used only for that purpose (on this point, see also Cangemi & Grice, 2016). In addition to following the AM approach to intonational phonology and meaning, we situate our work within an interactionist approach to communication, which treats meaning as being co-

constructed by speaker and addressee (cf. Gunlogson, 2003; Elder, 2019; Elder & Beaver, 2022).

Our investigation builds on our earlier research on the intonation of Greek wh-questions (Gryllia, Baltazani, & Arvaniti, 2018, 2019; Baltazani, Gryllia, & Arvaniti, 2020). Our earlier conclusions were based on laboratory speech (Baltazani et al., 2020; Gryllia et al., 2018) and perception experiments that tapped into the meaning of the two most frequent tunes, L*+H L-H% and L+H* L-L% (Baltazani et al., 2020; Gryllia et al., 2019). Here, we relied on unscripted speech and considered additional tunes and their relation to non-canonical wh-questions. Our purpose was threefold: first, to illustrate the range of tunes used with wh-questions in Greek; second, to discuss the pragmatic import of the tunes and provide a compositional analysis of them; third, to present an inventory of possible uses of wh-questions in Greek.

In brief, we show that wh-questions in Greek are produced with a variety of tunes, each of which is used both for different types of non-canonical wh-questions but also employed for other purposes. Therefore, each tune's meaning depends on and is augmented by lexical choices, context, and the interlocutors' knowledge of the situation. In turn, such knowledge or lack thereof can lead to different types of interactions (cf. Elder, 2019). Further, our pragmatic analysis of the tunes indicates that functions such as information structure or epistemic stance may be only *partially* encoded using intonation. Finally, the investigation of our corpus testifies to the need to examine spontaneous speech elicited in a variety of communicative situations, as these encourage the effortless use of different discourse devices. It follows that firm conclusions cannot be drawn either by relying on laboratory speech alone or by considering data elicited by means of only one task, however natural it may be.

In the remainder of the paper, we present our corpus (section 2), and the main tunes used with wh-questions in Greek together with their pragmatic interpretation (section 3). We then present the pragmatic uses of each main tune as attested in our corpus (section 4). Section 5 discusses the relationship between wh-questions and irony, while section 6 concludes.

2. The corpus

Our corpus comes from two sources, unscripted speech elicited for a phonetic study of intonation using three tasks, and unscripted (i.e. not read from a teleprompter) speech from audio recordings of radio, television and online shows.

The phonetic study corpus was elicited from 36 speakers of Greek (19 Female, 17 Male; mean age = 23.7 y.o.; SD = 3.2), all functional monolinguals (with the exception of one Greek-Albanian bilingual speaker). The speakers were recorded either in quiet locations during COVID (N = 8; 4F; mean age = 24.1 y.o.; SD = 1.7) or in the studio of the

Department of Musicology at the University of Athens (N = 28; 15F; mean age = 23.5 y.o.; SD = 3.6). The dataset used for the present study included wh-questions from three unscripted tasks: a) two rounds of Map Task (Anderson et al., 1984; Anderson et al., 1991), using different maps in each round with participants switching roles as instructor in one round and follower in the other; b) discussion, in pairs, of a set of unusual objects (such as an egg separator resembling a spider) and their possible uses (cf. Edlund et al., 2010); c) a game of "Who Am I" played in groups of four and using either a phone app (COVID group) or the board game *Μάντεψε τι + ποιος* "Guess what & who".

The media-based corpus consisted of audio recordings of cooking shows by three Greek celebrity chefs, Akis Petredzikis (Petredzikis, 2023, 2024), Argiro Barbarigou (Barbarigou, 2021), and Vefa Alexiadou (Alexiadou, 2024), and four episodes of Ellinofreneia (Ellinofreneia Official, 2024a, b, c, d, e), a radio show satirizing Greek politics. The cooking shows were selected because we expected the chefs to use pedagogical questions. We selected Ellinofreneia (a blend of *Hellas* and *schizophrenia*) because it is a political show that takes an ironic stance not only towards Greek politicians but also towards its own audience members who call in and engage in dialogue with the presenters. Thus, we expected to find rhetorical and ironic questions in this show. In total, the corpus consisted of approximately 500 wh-questions.

To analyze the corpus, we listened to the audio files, identified wh-questions and categorized them according to function and tune. With respect to function, we separated canonical from non-canonical questions and for the latter, we followed the classification of Eckardt (this volume). With respect to the tune, we focused on two main elements: the type of accent on the wh-word (which is fronted and focused), and the type of boundary tone on the question's right edge. Each author classified the questions in a subset of the corpus, with the final categorization dependent on discussion of these classifications and consensus among the authors; the rest of the corpus (N = xxx) was classified by an assistant based on this initial classification scheme, and checked by the second author The audio files used for the examples and figures in the paper (and some additional examples) can be found at: https://osf.io/qvw2g/.

3. Main tunes and pragmatic analysis

3.1 Main tunes

By default, Greek wh-questions are produced with a fronted wh-word that carries the nuclear pitch accent of the question (Alexopoulou & Baltazani, 2012). This pitch accent is either a L*+H or a L+H* (Baltazani et al., 2020). Phonetically, L*+H shows a rise from a low F0 point to a delayed peak, while L+H* shows a shallow rise with the peak aligned with the stressed vowel of the wh-word (Gryllia et al. 1018; Baltazani et al., 2020). See Figures 1 and 3 for examples of the two accent types.

The accentual rise is substantially curtailed under extreme tonal crowding, which is frequent, since most Greek wh-words start with a stressed syllable with a voiceless onset; e.g., [ti] “what”, [pu] “where”, [pos] “how”, [pços] “who”. Because of the potentially missing rise, in the present analysis, we used early vs. late peak alignment as our primary criterion for distinguishing between L+H* (early peak) and L*+H (late peak).

The accentual peak of both L*+H and L+H* is followed by a fall. F0 reaches the bottom of the speaker’s range and remains low until either the end of the question or the last stressed vowel, at which point F0 starts rising (Arvaniti & Ladd, 2009). Following Baltazani et al. (2020), we analyze the fall that persists to the end of the question as a sequence of L-L% edge tones, and the low F0 stretch that is followed by a rise as L-H%.

The above representations of the nuclear pitch accent and following edge tones give rise to four combinations: L*+H L-H%, L*+H L-L%, L+H* L-L%, and L+H* L-H%. We discuss the pragmatics of each of these tunes next.

3.2. A compositional analysis if the tune pragmatics

A unified account of the meaning of L*+H L-H%, L+H* L-L%, L+H* L-H%, and L*+H L-L% is provided based on Baltazani et al. (2020). Such a unified account is needed because, as mentioned, each tune is used for a variety of purposes: no tune is exclusively associated with just one communicative function, and the range of uses is such that it is not even possible to assume a distinction between tunes for canonical and tunes for non-canonical questions.

In Baltazani et al. (2020) we proposed that the suitability of each tune for a given context depends on the choice of pitch accent and boundary tone, with the two working either synergistically or antagonistically. The default tune for canonical questions, which is L*+H L-H%, illustrates this point: L*+H, the pitch accent typically used in Greek to accent words in prenuclear position, serves to highlight the wh-word without biasing the question. It further ensures that the tune is well-formed, as it fulfills the requirement that each utterance has at least one accent. The H% boundary tone opens the question to the addressee, indicating that an answer is expected. Together, the L*+H pitch accent and H% boundary tone indicate that the answer is to be chosen from an open set of possible answers and that the speaker assumes the answer is potentially known to the addressee. Thus, this tune is suitable for canonical questions, i.e., those that express genuine requests for information from the addressee. Henceforth we refer to these as *information-seeking questions*.

L+H* L-L% has a L+H* accent which is used in Greek to mark narrow focus (Arvaniti & Baltazani, 2005) and updates the information in the common ground. In wh-questions, its use indicates that the update in the answer is to be chosen from a closed set of potential answers (Rooth, 1992). Since the L% boundary tone indicates that no answer is expected (L% being used in statements as well), the combination of L+H* with L% generates certain implicatures: either there is no good answer to the question (the set of potential answers is null), or no answer is needed, because the answer is known to all and the speaker simply seeks the addressee's commitment with respect to the underlying proposition (Biezma & Rawlins, 2016; Bartels, 1997).[1] The former interpretation makes the tune suitable for biased questions, while the latter makes it suitable for rhetorical questions in the sense of Caponigro & Sprouse (2007), i.e., questions to which the answer is already known. However, as we discuss in more detail in section 4.1, this tune can be used for canonical questions as well.

In the other two tunes, L+H* L-H% and L*+H L-L%, the pitch accent and boundary tone operate antagonistically. L+H* L-H% indicates that an answer is expected from the addressee (because of the H%), while the use of L+H* indicates that the answer is to be chosen from a closed set of alternatives, presumably known to the speaker, and assumed to be known or easily inferable by the addressee. This can lead to a biased question with either positive or negative bias depending on the situation. For instance, a question such as [ˈti θa ˈfame ˈsimera] "what will we eat today?", when uttered with L+H* L-H%, is best translated in English as *guess what's for dinner?* where *guess* is conveyed by the tune. If such a question is used by a parent to a child, the small set of options is likely to be interpreted as the set of dishes the child particularly likes. However, the same question uttered by an antagonistic sibling could lead the same child to interpret the closed set of alternatives as one that includes the dishes they particularly dislike.

According to the analysis we have laid out, the final tune, L*+H L-L%, would be interpreted as follows: the answer is chosen from an open set, because of the L*+H accent, but the L% indicates that no answer is required. Thus, this tune may be suitable for formulaic questions such as [ˈpços ˈkseri] "who knows?" or [ˈpu aˈkustice] "who has heard of such a thing?", [ce ˈti na ˈkano eˈɣo ˈtora] "and what should I do now?" meaning *how is that my concern?* or [ˈti ˈpai na ˈpi] "so what?". In such questions the speaker indicates that they believe there to be a set of potential answers, but they are simultaneously declaring they are not interested in an answer.

[1] This interpretation of the L% boundary tone may seem inconsistent with the default polar question tune of Greek, L* H-L% (Arvaniti & Baltazani, 2005). A plausible explanation is that the difference is related to the phrase accent, L- in statements and wh-questions, but H- in polar questions. This suggests that the phrase accent may contribute to the meaning independently (Pierrehumbert & Hirschberg 1990 and Bartels 1997 on English) or as a unit with the boundary tone (cf. Portes & Beyssade, 2015). A discussion of these alternatives is beyond the scope of this paper.

In section 4 we investigate the canonical and non-canonical functions of the four tunes as they were appear in our corpus and show that the attested uses fit the above compositional analysis of intonation meaning.

4. Types of wh-questions in Greek

4.1 Canonical questions

Canonical, information-seeking questions are typically produced with either L*+H L-H% or L+H* L-L% (Arvaniti & Baltazani, 2005; Arvaniti & Ladd, 2009). As noted, L*+H L-H% is the best suited for this purpose. The use of L+H* L-L% for information-seeking questions, on the other hand, indicates that intonation alone cannot determine the interpretation of an utterance. However, since wh-questions are overtly marked in Greek, variation in tune with the same neutral intent is expected. The tunes are illustrated in Figures 1 and 2.

Baltazani et al. (2020) investigated the differences between the L-H% and L-L% ending tunes in wh-questions and found that participants rated the former as more polite than the latter. In terms of frequency, the L-H% tune appears to be more frequent and used by default when no instructions are given (Arvaniti & Ladd, 2009). Our present corpus confirms that the L*+H* L-H% tune is the default for information-seeking questions. It is illustrated in Figure 1 which comes from a Map Task, with the follower asking the instructor what they can see at a specific location on their map.

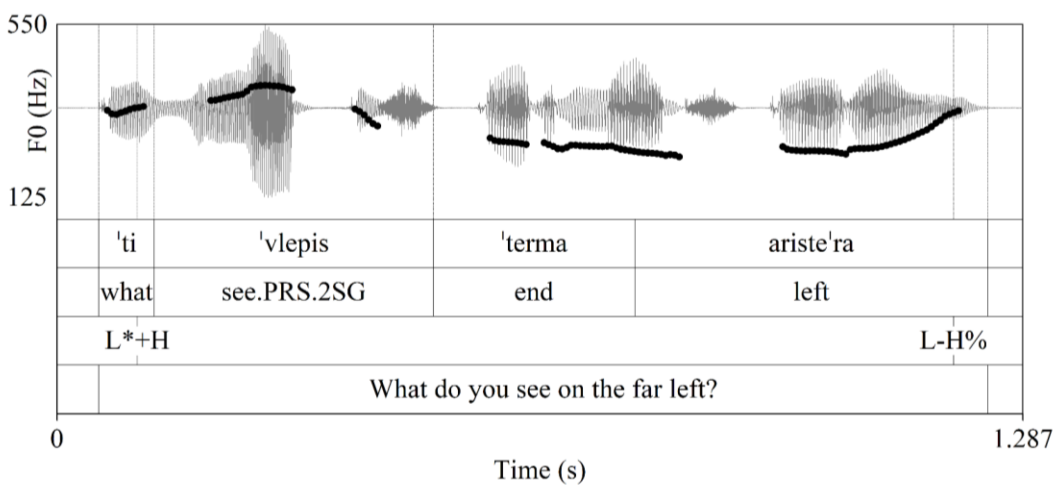

Figure 1. An information-seeking question with the L*+H L-H% tune; from a Map Task.

The follower question in line 7 of example (1), shown in bold and illustrated in Figure 2, showcases the use of the L+H* L-L% tune for unbiased information-seeking questions. In this dialogue, the two participants realize that there is a discrepancy in their maps: the instructor’s map includes an anemone, while the follower’s does not. Once the issue is resolved, the follower asks a clarification question using the L+H* L-L% tune; her utterance is treated as a straightforward information-seeking question with the instructor providing the requested information.

(1)

1	Follower	[ðen ˈexo aneˈmoni] “I don’t have an anemone.”
2	Instructor	[a'fto to… lu'luði] “That…flower.”
3	Follower	[ðen ˈexo ˈcapça aneˈmoni || aˈla | e | eˈci pu ˈeçis ta leˈmoɲa (pause) ˈine sta aristeˈra] “I don’t have an anemone. But, er, the spot where you have the lemons [...] is it on the left?”
4	Instructor	**['ti || 'ti enoˈis]** | /\ | /\ L+H* L-L% L+H* L-L% What what mean.PRS.2SG “What? What do you mean?”
5	Follower	[ˈeçis tin iˈkona me ta leˈmoɲa (pause) broˈsta su || sto aristeˈro su ˈçeri ˈine i aneˈmoni] “You have the image of the lemons [...] in front of you. Is the anemone on your left hand?”
6	Instructor	[ne] “Yes.”
7	Follower	[oˈrea || **c ap tin aneˈmoni | aˈpo pu ˈprepi na peˈraso**] “Great. And after the anemone, where should I go?”
8	Instructor	[θa peˈrasis aˈnamesa ap tin aneˈmoni ce ta ˈorima leˈmoɲa] “You will pass between the anemone and the ripe lemons.”

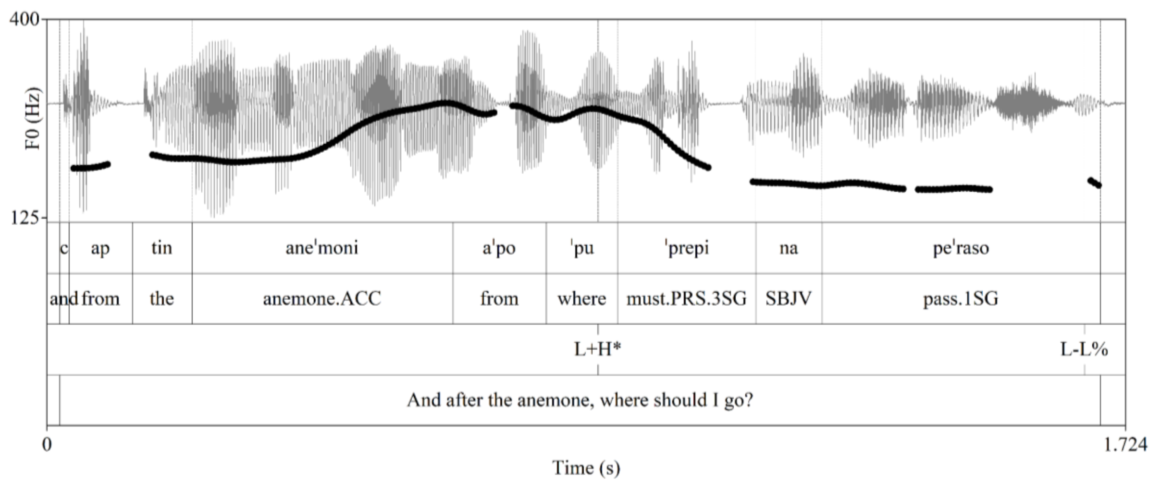

Figure 2. An information-seeking question with the L+H* L-L% tune; see (1) for the full dialogue excerpt; from a Map Task.

4.2 Biased questions

As discussed in section 3.2, L+H* L-L% is suitable for biased questions and is the tune used for this purpose in our corpus. Biased questions usually invite answers of the opposite polarity; informally, a positive polarity biased question usually implies a negative answer. For example, one can say to their partner [ˈpote maˈɟirepses teleˈftea foˈra] "When did you last cook?" using this tune to imply that the partner hasn't cooked for a long time. Such biased questions of positive polarity can function as indirectly stating something negative, including criticism. In such uses, in Greek, the addressee has three options: first, they can treat the utterance as a canonical information-seeking question and respond by providing information (e.g., the partner may reply *I cooked last Monday*); b) they may address only the negative implicatures generated by the tune (e.g., *You're right, I don't cook often enough*); c) finally, they may respond with a combination of the above, e.g. by saying *You're exaggerating, I did cook last Monday* (cf. Baltazani et al., 2020). The interpretation of the tune and response choice depend at least in part on previous knowledge (e.g., on whether cooking dinner is a regular dispute) or, depending on the situation, on general world knowledge. Consequently, whether the question with L+H* L-L% is treated as neutral or biased can depend on the circumstances.

Examples of biased questions from our corpus include the instructor question in line 4 of example (1), and the follower questions in (2) and (3); the question in (3) is illustrated in Figure 3. As already mentioned, in (1), the instructor and follower in a Map Task session needed to resolve the difficulty posed by a difference in their maps; when the follower tried to state the problem and asked a polar question, the instructor responded with two questions of his own, "What? What do you mean?", both with the L+H* L-L% tune. The follower, recognizing the negative implicature generated by the questions (something along the lines of *you are not being clear*), responded not by providing information but by asking a clarification question instead. Similarly, in the Map Task session in (2), the follower acknowledged the instruction, then referred to a previous instruction and finished her turn asking a wh-question. By using the L+H* L-L% tune, she implied that she could not go the way the instructor had told her to. As in (1), the instructor recognized the negative implicature and responded by asking a clarification question rather than providing information.

(2)

Instructor	[sto iˈɣro liˈvaði pas pros to ˈplai ɟa na peˈrasis aˈnamesa sto iˈɣro liˈvaði ce to ɟelaˈðari] “At the wetlands you go to the side so as to pass in between the wetlands and the cattle egret.”
Follower	[ne || ce meˈta mu ˈipes na ˈpao pros ta ˈkato] “Yes. And then you told me to go downwards.” **[ˈpu na ˈpao pros ta ˈkato]** | /\\ L+H* L-L% where SBJV go.PRS.1SG towards the down “Downwards in what direction?”
Instructor	[to ɟelaˈðari ðen ton ˈeçis aˈpenadi ap to iˈɣro liˈvaði] ‘Don’t you have the cattle egret across from the wetlands?’

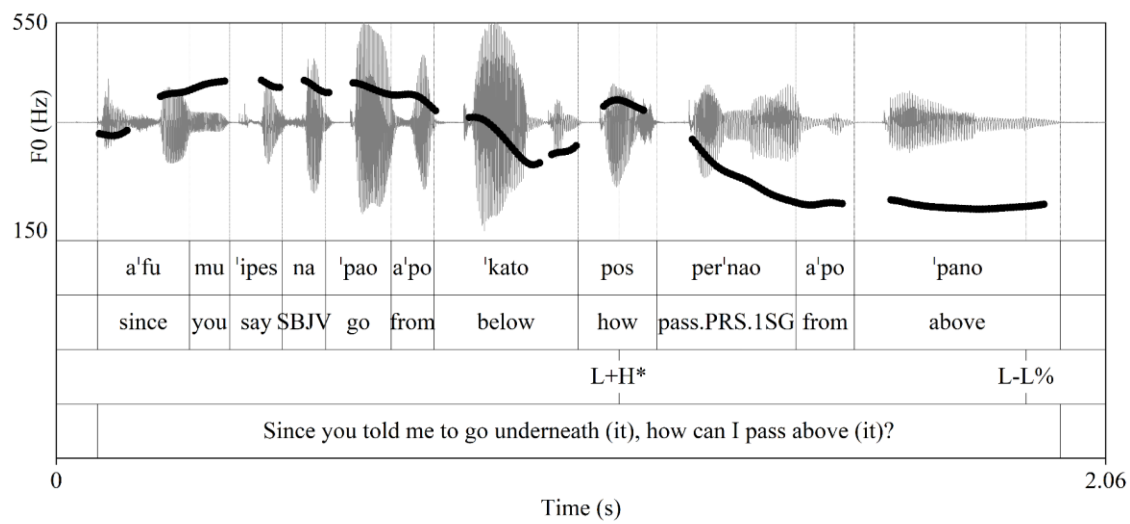

Figure 3. A biased question with the L+H* L-L% tune; see (3) for the full dialogue excerpt; from a Map Task.

(3)

Instructor	[perˈnas ˈpano ap to iˈɣro liˈvaði] “You go above the wetlands…”
Follower	[aˈfu mu ˈipes] [na ˈpao aˈpo ˈkato ||ˈ**pos perˈnao aˈpo ˈpano**] | /\\ L+H* L-L% (interrupts) “since you told me to go underneath (it), **how can I pass above (it)?”** (laughter)
Instructor	[ˈzvisto | ˈzvisto] “Rub it off, rub it off.”

4.3. Rhetorical questions

As noted in 3.2, L+H* L-L% is also suitable for rhetorical questions. These do not require an answer and can function as assertions (Hill & Miyagawa, 2024) because their answer is considered obvious to all interlocutors (Rohde, 2006; Caponigro & Sprouse, 2007; Biezma & Rawlins, 2016; Ferin, this volume). Such a definition acknowledges uses where the answer suggested by the rhetorical question is a non-empty set. An example of such a rhetorical question is given in (4). It comes from Ellinofreneia (Ellinofreneia, 2024b) and the topic is weight gain during the Christmas holidays; after one of the presenters suggests it is time to go on a diet, his interlocutor responds with “who starts a diet on a Thursday?”. The wh-question is clearly meant as rhetorical, as it is a recurring joke that diets start on a Monday.

(4) [ˈpemti | ˈðieta || ˈpços kseki ˈnai]
L+H* L-L%
Thursday.ACC diet.ACC who.NOM start.PRS.3SG
“Who starts a diet on a Thursday?”

Additionally, in our corpus we find rhetorical questions produced with L*+H L-L%. This tune is illustrated in Figure 4 which shows a rhetorical question by an Ellinofreneia presenter asking “why shouldn’t New Democracy [Greece’s ruling party] be re-elected?”. The answer to a rhetorical question that includes negation is positive; in this instance, then, the answer is *there is no reason why New Democracy shouldn’t be re-elected*. In this context, the rhetorical question is used ironically to imply that there are many reasons why New Democracy should not be re-elected. The use of this tune implies that the presenter does not really want an answer and thus that his intended message is in fact the opposite of what the words of his question are saying. Understanding the irony requires real world knowledge, in this case, knowledge of the presenter’s anti-New Democracy stance.

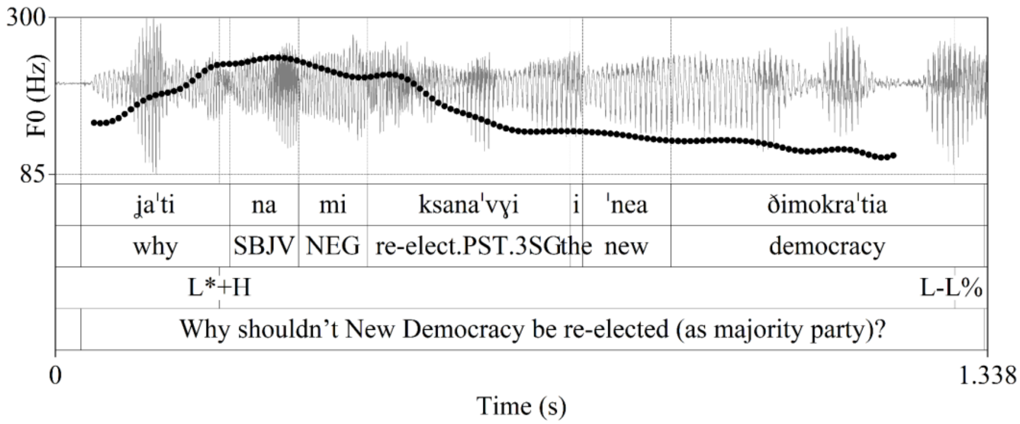

Figure 4. A rhetorical question with the L*+H L-L% tune; from Ellinofreneia (2024a).

4.4 Pedagogical and theme-setting questions

L+H* L-H% is largely used for pedagogical and theme-setting questions. In some contexts, such questions can be biased, though the bias is only understood if the addressee can make a reasonable guess of the speaker's intentions, as illustrated with the example of [ˈti θa ˈfame ˈsimera] "(guess) what are we having for dinner?" discussed in section 3.2.

In our corpus, we did not find theme-setting questions, though as native speakers we believe these are possible in Greek and, if used, they would be produced with the L+H* L-H% tune. For example, the presenter of a cooking or a craft show can start with a question such as [ˈti θa ˈftçaksume simera | ˈfiles ce ˈfili] "what will we make today, friends?".

In our corpus, the L+H* L-H% tune was used extensively with pedagogical questions. These were very frequent in the cooking shows, where the presenter addressed either someone among the crew or the audience directly. This use is a rhetorical device (*anthypophora*) which is frequent in Greek: wh- as well as polar questions are used as a way to draw attention or steer the discussion in a specific direction. An answer is required, and the use of L+H* implies that the set of answers is closed.

From a prosodic perspective, most of the pedagogical questions with L+H* L-H% are characterized by a pitch span upward expansion for the H%. The expansion is often so extensive that the H% boundary tone is scaled noticeably higher than the H tone of the L+H* pitch accent; this scaling difference is the reverse of what is typical with this tune. It is unclear at this point whether this is a meaningful difference between high- and low-scaled H% boundary tones, though the most likely interpretation is that the pitch span expansion is paralinguistic; i.e., it is there to draw attention, a function in line with the speaking style of the type of discourse where most of these questions appear.

The pedagogical use of the L+H* L-H% is illustrated in Figure 5. In this example, chef Petredzikis is explaining the process of making rice. To explain a specific step, turning the heat down as soon as the water reaches boiling point, he turns this piece of information into a question that he addresses to his cameraman: “As soon as I see the first bubble, what do I do, Dionissi?”. Petredzikis then immediately answers the question himself with *I turn it down to the minimum*.

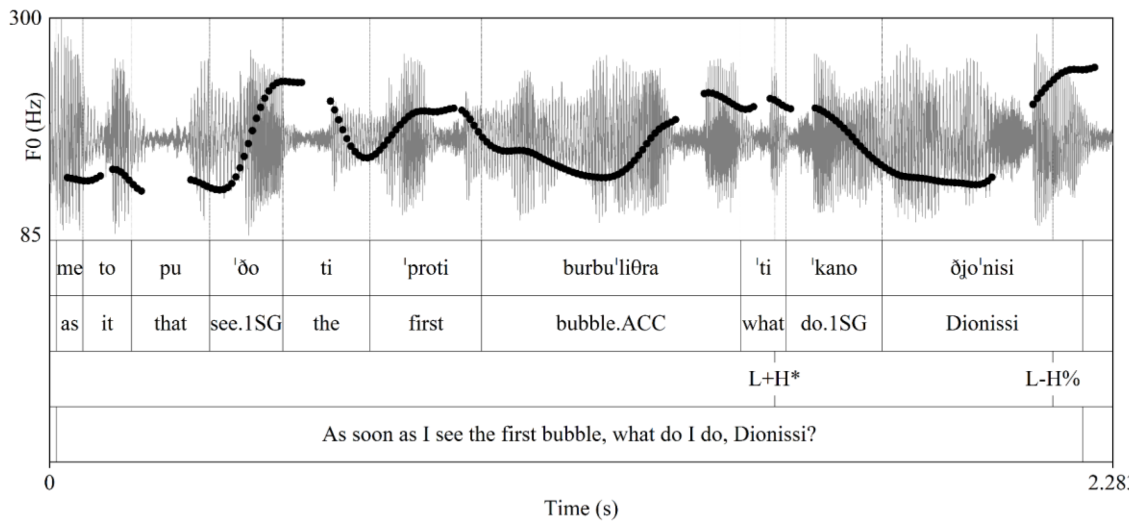

Figure 5. A pedagogical question with the L+H* L-H% tune; from Petredzikis (2024).

Pedagogical questions can be answered either by the speaker, as in the above example, or by the addressee. In our corpus we founnd both options. For instance, chef Petredzikis’ answers most of his own questions, but on occasion, the cameraman offers an answer that is accepted as correct and sometimes augmented with additional information provided by the chef. In one such exchange, Petredzikis gives instructions for vegan chicken nuggets by saying: [ˈbenune sto ˈfurno | ce ˈposi ˈora xriˈazode ˈfile ðʝoˈnisi] “they go into the oven, and how long do they need, Dionisis my friend?” The cameraman responds with the cooking time (10-12 minutes) which then Petredzikis repeats: *ten to twelve minutes, ok?* (Petredzikis, 2024).

Though pedagogical questions are particularly frequent in the cooking shows, they are not absent from other types of discourse we considered and were also typically answered by the speaker themselves. For instance, in Ellinofreneia (2024c), a caller is talking about the inability of Greeks to pay their bills by asking: “why is a person in debt?” and continues herself with the answer: *because they have been devastated by taxation*.

4.5. Conjectural questions

Conjectural questions – sometimes called ‘engaging’ or ‘inclusively self-addressed’ questions (Farkas, 2020) – express the speaker’s curiosity or speculation about a certain issue, rather than directly requesting an answer from the listener (Eckardt & Beltrama, 2019). An example is given in Figure 6, in which two speakers discuss the possible use of an unusual object and wonder what it might be. In this context, it is mutually assumed that the answer will be reached after a series of collaborative moves involving both conversation participants. Conjectural questions can have any of the tunes discussed here, L*+H L-H%, L+H* L-L%, L+H* L-H%, L*+H L-L%. This is illustrated in Figures 6 and 7, in which conjectural questions are produced with L*+H L-H% and L*+H L-L% respectively, while in example (4) the tune in is L+H* L-L%.

Conjectural questions are often combined with particles of epistemic modality and other discourse markers and intensifiers that may indicate more uncertainty on the part of the speaker (Eckardt & Beltrama, 2019). The option of using such markers distinguishes conjectural from information-seeking questions. For example, the question in (4) displays several such markers, the use of [boˈri] “may” which is obligatorily followed by the subjective [na xrisiˈmevi] “be used”, and the use of the particle [ˈaraɟe] loosely translated as “I wonder”. Note that none of these elements are suitable for an information-seeking question, which would be [ˈpu xrisiˈmevi] “What purpose does it serve?”. The question in Figure 6 also uses [boˈri] followed by the verb in the subjective, while Figure 7 shows additional markers, the use of the subjunctive ([na xrisiˈmevi] “be used”), “but” ([ˈomos]), and the interjection [re peˈði mu] loosely translated as “you guys”.

(4) [ˈpu boˈri na xrisiˈmevi ˈaraɟe]
| /\
L+H* L-L%
where may.PRS.3SG SBJV be.of.use.PRS.3SG particle
“What purpose might it serve, I wonder?”

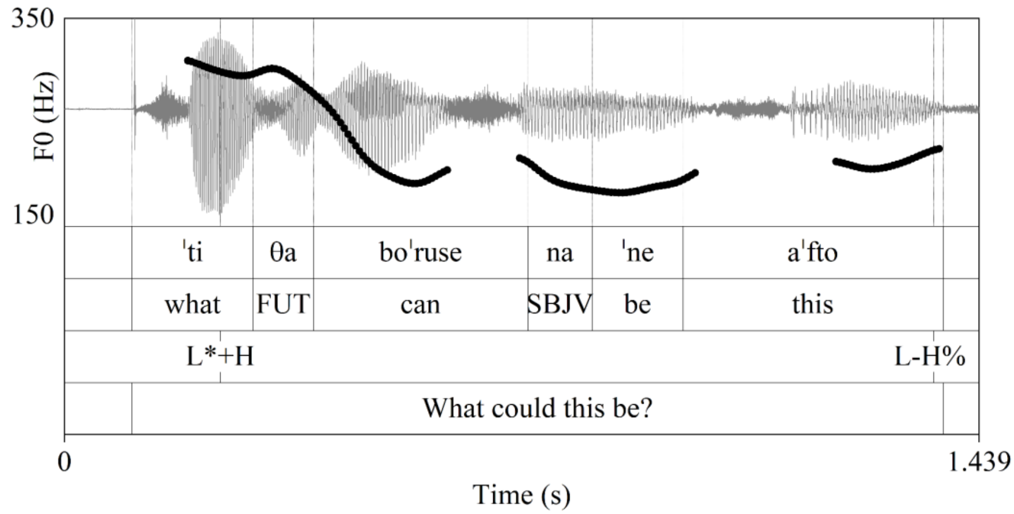

Figure 6. A conjectural question from an objects dialogue, with the L*H L-H% tune.

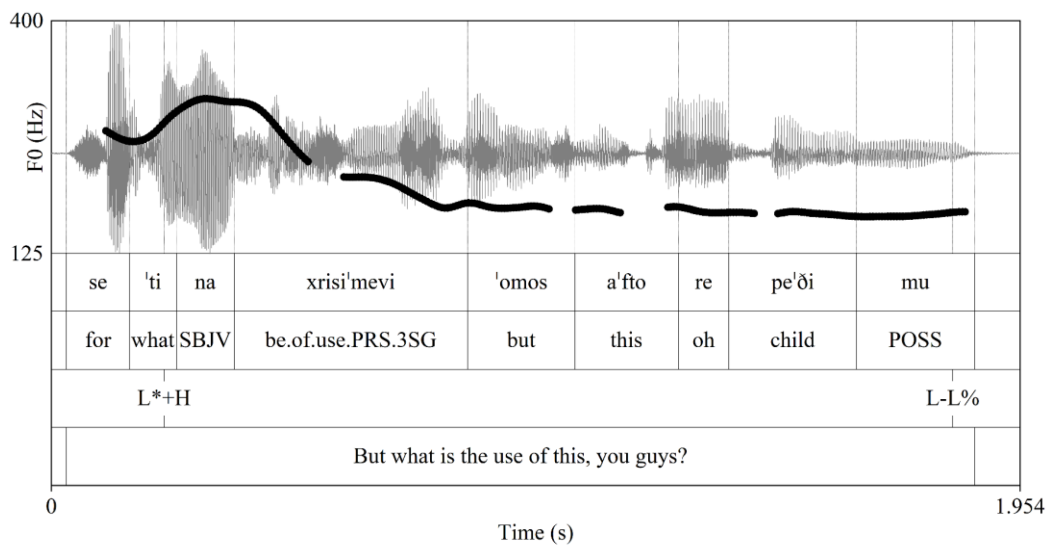

Figure 7. A conjectural question from an objects dialogue, with the L*H L-L% tune.

5. Wh-questions and irony

In our corpus, wh-questions were frequently used for irony, particularly in Ellinofreneia. Most of the ironic questions in our corpus were used to express the opposite of what the tune would normally imply (e.g., Kapogianni, 2021), whether this means that the question is also to be interpreted as information-seeking, biased, or conjectural. Consequently, ironic questions can be uttered with any tune, depending on how irony is expressed. This is illustrated in Figures 4, 8, and 9.

In Figure 4, the question has the L+H* L-L% tune, which is prima facie the best choice for irony: the answer set is null or limited and no answer is necessary. Figure 8, however, shows that this is not the only option. In this instance, a caller in Ellinofreneia (Ellinofreneia, 2024c) starts talking about *Papadopoulos*, the leader of Greece's brutal military junta of 1967-1974. Such mentions often indicate the speaker views this period as one of order and stability. To subvert this reading, the presenter interrupts with a question, "which Papadopoulos?", with the typical information-seeking tune, L*+H L-H%. By using this tune the presenter feigns ignorance of which person the caller is referring to. Indirectly, this move – asking for information – allows the presenter to cancel the claimed connection between the dictator Papadopoulos and peace and safety and thus forces an implicit acknowledgement of who *Papadopoulos* really was, a dictator. This example nicely demonstrates that the information-seeking tune can also convey irony, given an appropriate context.

Finally, in Figure 9, the tune is L+H* L-H%. The topic is, once more, the uselessness of Greek elections (Ellinofreneia, 2024b). Here the tune typical of conjectural questions is employed, though there is no forthcoming answer. The presenter uses the tune ironically to imply the opposite of what his utterance ostensibly expresses: though he speculates that voters need not engage in the election as it is futile (*why bother?*), the audience can grasp the underlying message that elections are indeed essential

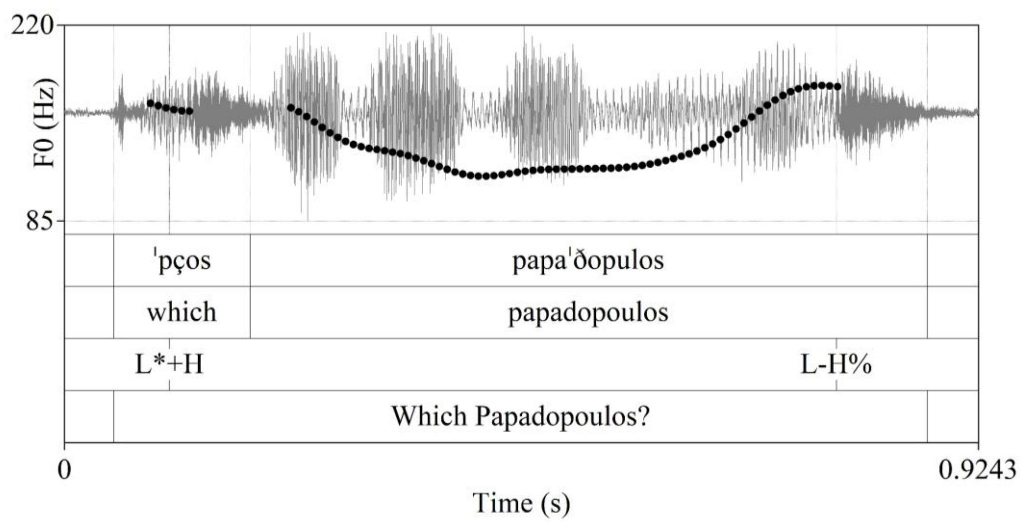

Figure 8. A question with the L*+H L-H% tune, typical for information-seeking questions, used ironically; from (Ellinofreneia, 2024c)

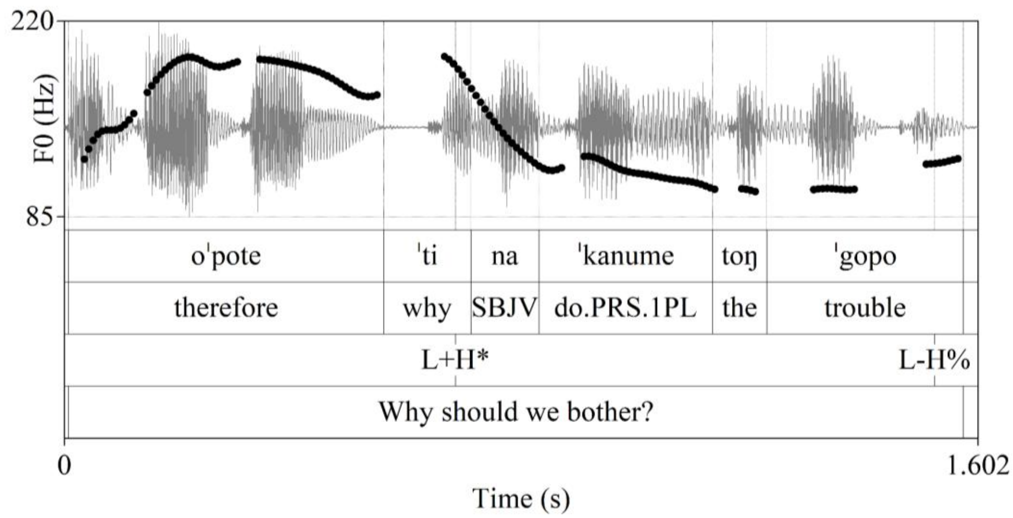

Figure 9. A question with the L+H* L-H% tune, leading to a biased question with an ironic interpretation; from (Ellinofreneia, 2024c)

6. Discussion and Conclusion

We discussed a number of tunes used in Greek with wh-questions, and offered both a prosodic and a pragmatic analysis for them. By examining a corpus of spontaneous uses of wh-questions in different communicative situations, we have shown that wh-questions serve multiple functions, all in line with Eckardt's categories (Eckardt, this volume).

The wh-questions in our corpus were produced with one of four tunes, L*+H L-H%, L+H* L-L%, L+H* L-H%, and L*+H L-L%, which are not exclusive to question forms but extend to other utterance types as well. Notably, at least two of these tunes are used with other utterance types. For example, L*+H L-H% is also used with certain negative statements (Baltazani 2002, 2006). Similarly, L+H* L-L% is used in declarative statements with contrastive or corrective focus (Arvaniti & Baltazani, 2005, and Arvaniti, Katsika, Hu, 2024). These patterns underscore the absence of one-to-one correspondence between tune and meaning: instead, the mapping is many-to-many.

In addition to the prosodic analysis of the tunes, we proposed a pragmatic account grounded in compositional approaches to intonation meaning, primarily based on Pierrehumbert & Hirschberg (1990), with additional insights adopted from Steedman (2014). Our main argument is that compositional models better capture the flexibility of tune interpretation than configurational ones, as they allow individual elements of a tune to retain abstract meanings across varied lexical and contextual environments. Intonational systems cannot be argued to encode specific functions—such as information structure, theme-rheme distinctions, or epistemic stance (Pierrehumbert & Hirschberg, 1990; Steedman, 2014; Bartels, 1997; Prieto & Borràs-Comes, 2018)—in a discrete or exclusive manner. These functions may be only partially signaled through intonation.

Lexical content and context play a pivotal role, shaping implicatures and enabling interpretations such as irony, which depends not only on the tune but also on shared world knowledge.

This complexity means that intonation cannot be used to neatly classify the functions of wh-questions. Many serve multiple functions simultaneously. The use of rhetorical questions as an ironic device is a case in point. Moreover, as noted in section 4.2 and already hinted at in Baltazani et al. (2020), biased questions may be interpreted primarily for their presupposed stance rather than as information-seeking acts, with interlocutors addressing the implied bias rather than providing new information.

The function of the question, therefore, is not always unique and unambiguous. On many occasions, interpretations required listening to the extended context in which our examples were situated and discussion among us to best determine the function of a particular question in a given context. This reflects the nature of the communication process, which is probabilistic and open to misinterpretation (Elder, 2019). These findings reinforce the view that examining the pragmatics of intonation without considering discourse context yields limited insight. Pragmatics frameworks rooted in interactional approaches (e.g., Ariel, 2016; Elder, 2019; Elder & Beaver, 2022) offer a more productive path forward.

This investigation highlights the need to use spontaneous data that come from different sources and serve different communicative functions. In a laboratory environment, speakers may consistently opt for a default tune, simply because the traditional tasks, such as reading sentences aloud, lack communicative intent. This tendency may be especially notable with less frequently used tunes, which are hard to elicit in the laboratory (Arvaniti, 2016); it may also be related to a specific speaker's ability to imagine the right context and use it to select an appropriate tune. This limitation of lab speech was particularly notable with respect to the L*+H L-L% tune. In Baltazani et al. (2020), we speculated about its meaning but did not have concrete examples of its use. The spontaneous speech examined here provided instances of L*+H L-L% and allowed us to better understand the functions of this tune.

Moreover, our analysis demonstrates that the limitations of laboratory speech cannot be overcome by eliciting data through some frequently used unscripted tasks: our data revealed variations in the usage of wh-questions depending on the task which largely dictates participant moves. As an example, rhetorical questions were frequent In Ellionophreneia, while pedagogical questions abounded in cooking shows, yet neither appeared in the map task sessions. The importance of analyzing spontaneous and unscripted speech from varied communicative situations is underscored by the discovery of instances of L*+H L-L%, which eluded us in laboratory settings and would have remained undetected had we restricted our corpus to a single task.

Our description of tunes, their functions, and pragmatics does not cover all our observations regarding wh-questions in our corpus. First, although our analysis focused on F0, the tunes involved additional phonetic detail, such as changes in pitch scaling, segmental duration (see also Ferin, this volume on syllable duration in Italian questions), and on occasion speaking rate (see also Seeliger, this volume, on speech rate in Swedish). Impressionistically, many pedagogical questions showed a fast speaking rate at the beginning coupled with the already mentioned expansion of pitch span for the final H% boundary tone. Variability may stem from factors such as politeness, which may slow speaking rate and raise high tones, and surprise, which may raise pitch level or increase span. It is also possible that features such as the exceptionally high scaling of H% and accelerated speaking rate with pedagogical questions could be integral to the realization of the L+H* L-H% tune in specific contexts. Though such changes seem to be largely stylistic rather than structural, they may constitute redundant features that help listeners interpret the tune and thus the pragmatic intent of the question (cf. Braun et al., 2019). Thus, a systematic overview of such features should more regularly accompany the description of tunes to shed light on their function.

In conclusion, we have provided a brief description of the prosody and uses of wh-questions in Greek, based on a large and varied corpus of unscripted speech. Our data indicate that wh-questions are produced with a variety of tunes, none exclusive to wh-questions, and that wh-questions serve a variety of pragmatic purposes in Greek. Our findings advocate for a nuanced compositional approach to intonation meaning based on a variety of spontaneous data, which is essential for advancing our understanding of both intonation structure and intonation meaning.

Acknowledgments

We thank Riccardo Orrico for his valuable comments to an earlier version of the manuscript, Christina Zygouri for cataloguing the questions in our corpus, Dafne Bagioka for data collection in Athens, Angelos Lengeris, Anastasia Georgaki, and Ioanni Peikidis for facilitating the Athens recordings, and Eleni Kapogianni for introducing us to Ellinofreneia. This work has been financially supported by the European Science Foundation through grant ERC-ADG-835263 (SPRINT) to Amalia Arvaniti, and by the British Academy through grant SG160538 to Amalia Arvaniti, Mary Baltazani & Stella Gryllia. This support is hereby gratefully acknowledged.

References

Alexiadou, V. (Director). (2024, January 18). Ψαροκεφτέδες της Βέφας Αλεξιάδου [Vefa Alexiadou’s fishballs]. https://www.youtube.com/watch?v=tTOkvLUepZs

Alexopoulou, T. & M. Baltazani. (2012). Focus in Greek wh-questions. In A. Neeleman & I. Kucerova (Eds.), *Information Structure: Contrasts and Positions* (pp. 206–246). Cambridge: Cambridge University Press.

Anderson, A. H., Bader, M., Bard, E. G., Boyle, E., Doherty, G., Garrod, S., Isard, S., Kowtko, J., McAllister, J., Miller, J., Sotillo, C., Thompson, H. S., & Weinert, R. (1991). The HCRC Map Task Corpus. *Language and Speech*, *34*(4), 351–366. https://doi.org/10.1177/002383099103400404

Ariel, M. (2016). Revisiting the typology of pragmatic interpretations. *Intercultural Pragmatics*, *13*, 1–35. https://doi.org/10.1515/ip-2016-0001

Arvaniti, A. (2016). Analytical decisions in intonation research and the role of representations: Lessons from Romani. *Laboratory Phonology*, *7*(1), 1–43. https://doi.org/10.5334/labphon.14

Arvaniti, A. (2022). The Autosegmental-Metrical model of intonational phonology. In S. Shattuck-Hufnagel & J. Barnes (Eds.), *Prosodic Theory and Practice* (pp. 25–84). Cambridge, Massachusetts: The MIT Press. https://doi.org/10.7551/mitpress/10413.003.0004

Arvaniti, A., & Baltazani, M. (2005). Intonational analysis and prosodic annotation of Greek spoken corpora. In S. Jun (Ed.), *Prosodic Typology: The Phonology of Intonation and Phrasing* (pp. 84–117). Oxford: Oxford University Press.

Arvaniti, A., Katsika, A., & Hu, N. (2024). Variability, overlap, and cue trading in intonation. *Language* 100(2): 265–307.

Arvaniti, A., & Ladd, D. R. (2009). Greek wh-questions and the phonology of intonation. *Phonology*, *26*(1), 43–74. https://doi.org/10.1017/S0952675709001717.

Baltazani, M. (2002). *Quantifier Scope and the Role of Intonation in Greek*. Unpublished Ph.D. dissertation, University of California, Los Angeles.

Baltazani, M. (2006). Intonation and pragmatic interpretation of negation in Greek. *Journal of Pragmatics*, *38*(10), 1658–1676. https://doi.org/10.1016/j.pragma.2005.03.016

Baltazani, M., Gryllia, S., & Arvaniti, A. (2020). The intonation and pragmatics of Greek wh-questions. *Language and Speech*, *63*(1), 56–94. https://doi.org/10.1177/0023830918823236

Barbarigou, A. (Director). (2021, May 21). Τσιζκέικ της Αργυρώς | Αργυρώ Μπαρμπαρίγου [Argyro's cheesecake]. https://www.youtube.com/watch?v=5rhunXLOSYs

Bartels, C. (1999). *The Intonation of English Statements and Questions: A Compositional Interpretation*. New York, London: Routledge.

Biezma, M., & Rawlins, K. (2016). Or what?: Challenging the speaker. In C. Hammerly & B. Prickett (Eds.), *NELS 46: Proceedings of the Forty-Sixth Annual Meeting of the North East Linguistic Society*, vol.*1*, (pp. 93–106).

Braun, B., Dehé, N., Neitsch, J., Wochner, D., & Zahner, K. (2019). The prosody of rhetorical and information-seeking questions in German. *Language and Speech*, *62*(4), 779–807. https://doi.org/10.1177/0023830918816351

Büring, D. (2009). Towards a typology of focus realization. In M. Zimmermann & C. Féry (Eds.), *Information Structure* (1st ed.), (pp. 177–205). Oxford: Oxford University Press. https://doi.org/10.1093/acprof:oso/9780199570959.003.0008

Caponigro, I., & Sprouse, J. (2007). Rhetorical questions as questions. *Proceedings of Sinn Und Bedeutung*, *11*, 121–133. https://doi.org/10.18148/sub/2007.v11i0.635

Eckardt, R. (this volume). The landscape of non-canonical questions.

Eckardt, R., & Beltrama, A. (2019). Evidentials and questions. In C. Pinon (Ed.) *Empirical Issues in Syntax and Semantics 12*, (pp. 121–155). http://www.cssp.cnrs.fr/eiss12/index_en.html

Edlund, J., Beskow, J., Elenius, K., Hellmer, K., Strönbergsson, S., & House, D. (2010). Spontal: A Swedish spontaneous dialogue corpus of audio, video and motion capture. *Proceedings of the Seventh International Conference on Language Resources and Evaluation (LREC'10)*. European Language Resources Association (ELRA). http://www.lrec-conf.org/proceedings/lrec2010/pdf/352_Paper.pdf

Elder, C.-H. (2019). Negotiating what is said in the face of miscommunication. In P. Stalmaszczyk (Ed.), *Philosophical Insights into Pragmatics* (pp. 107–126). Berlin, Boston: De Gruyter. https://doi.org/10.1515/9783110628937-006

Elder, C.-H., & Beaver, D. (2022). "We're running out of fuel": When does miscommunication go unrepaired? *Intercultural Pragmatics*, *19*(5), 541–570. https://doi.org/10.1515/ip-2022-5001

Ellinofreneia Official (Director). (2024a, January 07). Ελληνοφρένεια 07/01/2024 | https://www.youtube.com/watch?v=zZUxw-Aq2i0

Ellinofreneia Official (Director). (2024b, January 18). Ελληνοφρένεια 18/1/2024 | https://ellinofreneianet.gr/radio/radio-shows/40356-ellinofreneia-18-1-2024.html

Ellinofreneia Official (Director). (2024c, January 19). Ελληνοφρένεια 19/01/2024 | https://ellinofreneianet.gr/radio/radio-shows/40370-eellinofreneia-19-1-2024.html

Ellinofreneia Official (Director). (2024d, March 22). Ελληνοφρένεια 22/03/2024 | https://ellinofreneianet.gr/radio/radio-shows/41519-ellinofreneia-22-3-2024.html

Ellinofreneia Official (Director). (2024e, March 26). Ελληνοφρένεια 26/03/2024 | https://ellinofreneianet.gr/radio/radio-shows/41577-ellinofreneia-26-3-2024.html

Farkas, D. F. (2020). *Canonical and non canonical questions*. University of California at Santa Cruz. Available online: https://semanticsarchive. net/Archive/WU2ZjIwM/questions.pdf (accessed on 25 March 2024)

Ferin, M. (this volume). Prosodic patterns of Italian rhetorical questions.

Gryllia, S., Baltazani, M., & Arvaniti, A. (2018). *The role of pragmatics and politeness in explaining prosodic variability*. *Speech Prosody* (pp. 158–162). https://doi.org/10.21437/SpeechProsody.2018-32

Gryllia, S., Baltazani, M., & Arvaniti, A. (2019). Evidence for the compositionality of tunes and intonational meaning. In S. Calhoun, P. Escudero, M. Tabain & P. Warren (Eds.) *Proceedings of the 19th International Congress of Phonetic Sciences*, (pp. 2841–2845). Australasian Speech Science and Technology Association Inc., and International Phonetic Association. https://www.internationalphoneticassociation.org/icphs-proceedings/ICPhS2019/papers/ICPhS_2890.pdf

Gunlogson, C. (2003). *True to Form: Rising and Falling Declaratives as Questions in English*. New York, London: Routledge. ISBN 9780415865074.

t' Hart, J. T., Collier, R., & Cohen, A. (1990). *A Perceptual Study of Intonation: An Experimental-Phonetic Approach to Speech Melody*. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9780511627743

Hill, V., & Miyagawa, S. (2024). The commitment of rhetorical questions. *Glossa: A Journal of General Linguistics*, *9*(1). https://doi.org/10.16995/glossa.10360

Im, S., Cole, J., & Baumann, S. (2018). The probabilistic relationship between pitch accents and information status in public speech. *Speech Prosody 2018* (pp. 508–511). https://doi.org/10.21437/SpeechProsody.2018-103

Krifka, M. (2008). Basic notions of information structure. *Acta Linguistica Hungarica*, *55*(3–4), 243–276.

Kurumada, C., & Roettger, T. B. (2022). Thinking probabilistically in the study of intonational speech prosody. *Wiley Interdisciplinary Reviews: Cognitive Science*, 13(1), e1579. https://doi.org/10.1002/wcs.1579

Ladd, D. R. (2008). *Intonational Phonology* (2nd ed.). Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9780511808814

Petredzikis, Akis (Director). (2023, December 10). *Εύκολο Πανετόνε Επ. 18 | Kitchen Lab TV | Άκης Πετρετζίκης*. [Easy panettone] https://www.youtube.com/watch?v=ThOLGMFpeQg

Petredzikis, Akis (Director). (2024, January 20). *Vegan Poke Bowl με Nuggets Επ. 25 | Kitchen Lab TV | Άκης Πετρετζίκης*. [Vegan poke bowl with nuggets] https://www.youtube.com/watch?v=preF10mhgm0

Pierrehumbert, J. B. (1980). *The Phonology and Phonetics of English Intonation.* PhD Dissertation, Massachusetts Institute of Technology.

Pierrehumbert, J., & Hirschberg, J. (1990). The meaning of intonational contours in the interpretation of discourse. In P. R. Cohen, J. Morgan, & M. E. Pollack (Eds.), *Intentions in Communication* (pp. 271–312). The MIT Press. https://doi.org/10.7551/mitpress/3839.003.0016

Portes, C., & Beyssade, C. (2015). Is intonational meaning compositional? *Verbum*, *37* (2), 207–233.

Prieto, P., & Borràs-Comes, J. (2018). Question intonation contours as dynamic epistemic operators. *Natural Language & Linguistic Theory*, *36*(2), 563–586. https://doi.org/10.1007/s11049-017-9382-z

Rooth, M. (1992). A theory of focus interpretation. *Natural Language Semantics*, *1*(1), 75–116. https://doi.org/10.1007/BF02342617

Seeliger, H. (this volume). The prosody of Swedish non-canonical questions.

Steedman, M. (2000). Information structure and the syntax-phonology interface. *Linguistic Inquiry*, *31*(4), 649–689. https://doi.org/10.1162/002438900554505

Steedman, M. (2014). The surface-compositional semantics of English intonation. *Language*, *90*(1), 2–57. https://doi.org/10.1353/lan.2014.0010

Westera, M., Goodhue, D., & Gussenhoven, C. (2020). Meanings of tones and tunes. In C. Gussenhoven & A. Chen (Eds.), *The Oxford Handbook of Language Prosody* (pp. 442–453). Oxford: Oxford University Press. https://doi.org/10.1093/oxfordhb/9780198832232.013.29

Xu, Y. (2005). Speech melody as articulatorily implemented communicative functions. *Speech Communication, 46*, 220–251.